

DATA 260 Homework 1

Section 0. Personal Configuration and Domain

SID4	9486
PORT_BASE	8486
PREFIX	s9486
SEED	9486
VERIFY_SEED	269486
DOMAIN_ID	6
Assigned Domain	Rental housing listings
Hardware	Apple Mac with M1 chip and 8 GB unified memory
Local Model	qwen3:4b
Content-complete source commit hash	832703b
Final submission tag	hw1

The recommended qwen3:8b model was replaced with qwen3:4b because the available Mac hardware has only 8 GB of unified memory, so the smaller model is more practical for repeated local runs.

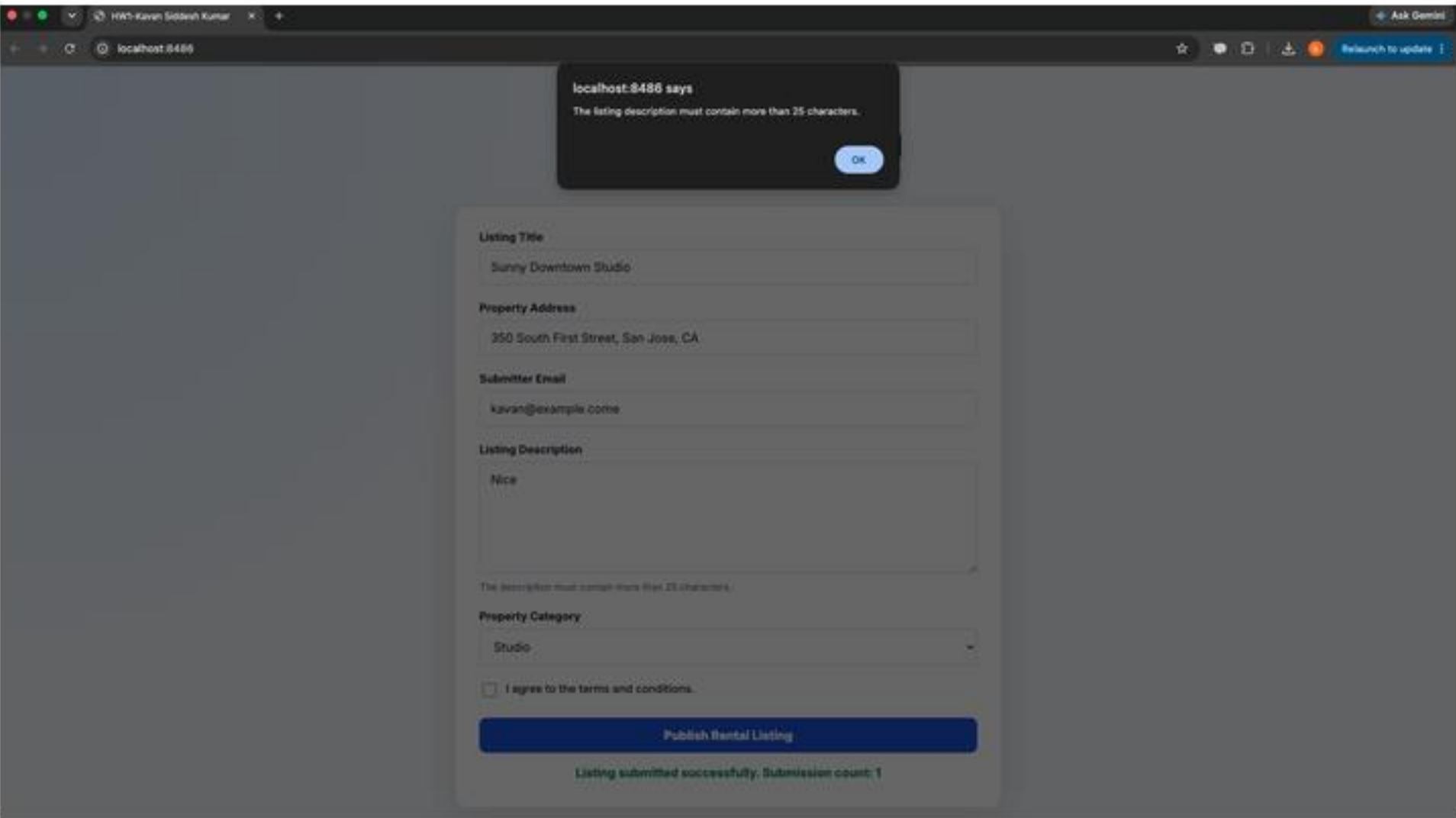
Repository notes: the shared adapter lives in `src/model_client.py`, the strict review prompt is in `AGENT.md`, and the final PDF is generated from the same evidence stored in `reports/hw01/`.

Reproducible commands

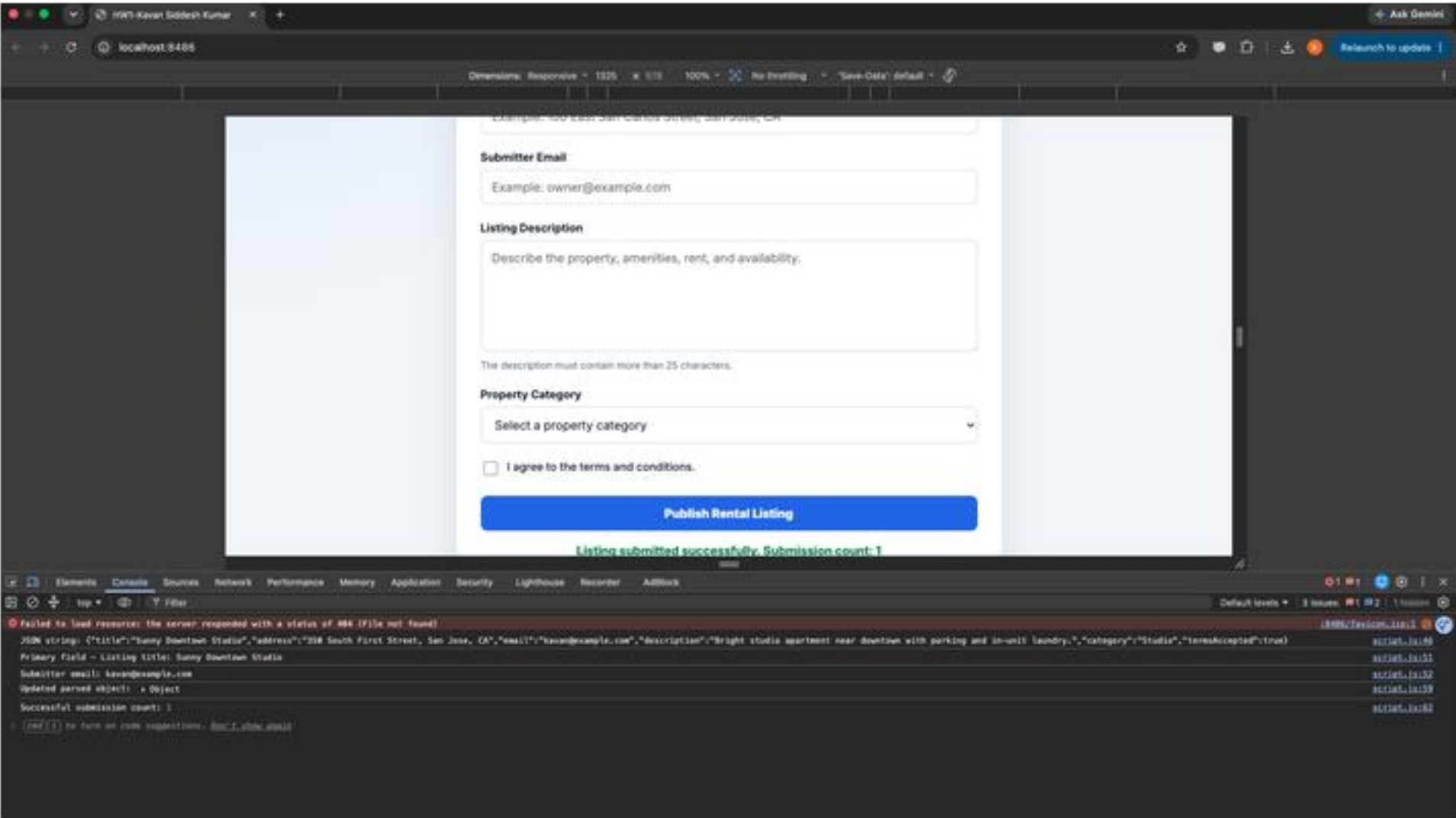
```
python3 -m http.server 8486 --directory app
docker build -t data260-hw1:s9486 .
docker run --name data260-hw1-s9486 -p 8486:80 data260-hw1:s9486
docker stop data260-hw1-s9486
docker rm data260-hw1-s9486
```


Section 1. Part 1: Rental Listing Application

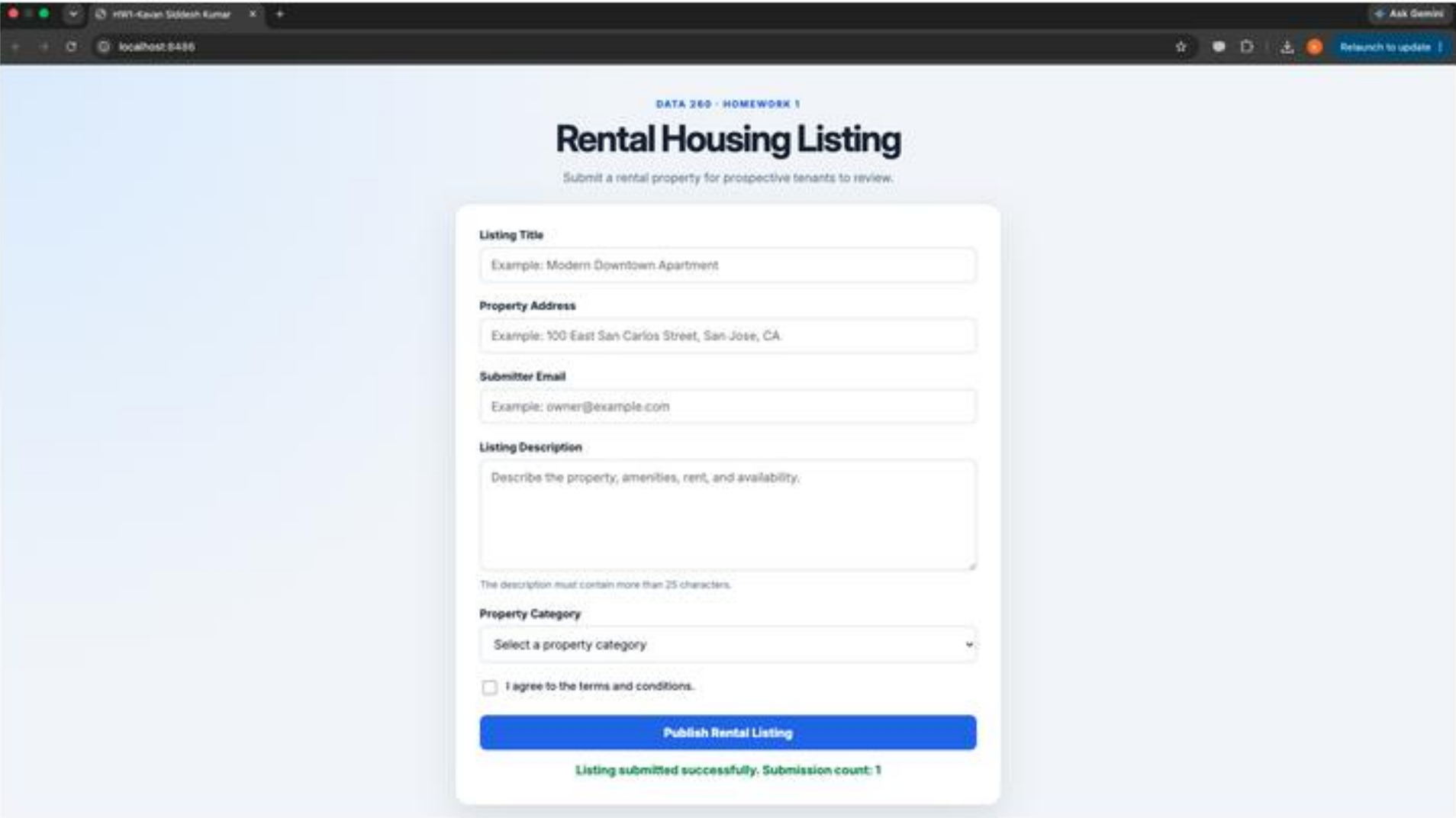
- The HTML form collects the domain fields from DOMAIN_SCHEMA.md.
- The JavaScript validates the description length and terms checkbox, serializes the listing, logs the important fields, and adds a submissionDate timestamp.



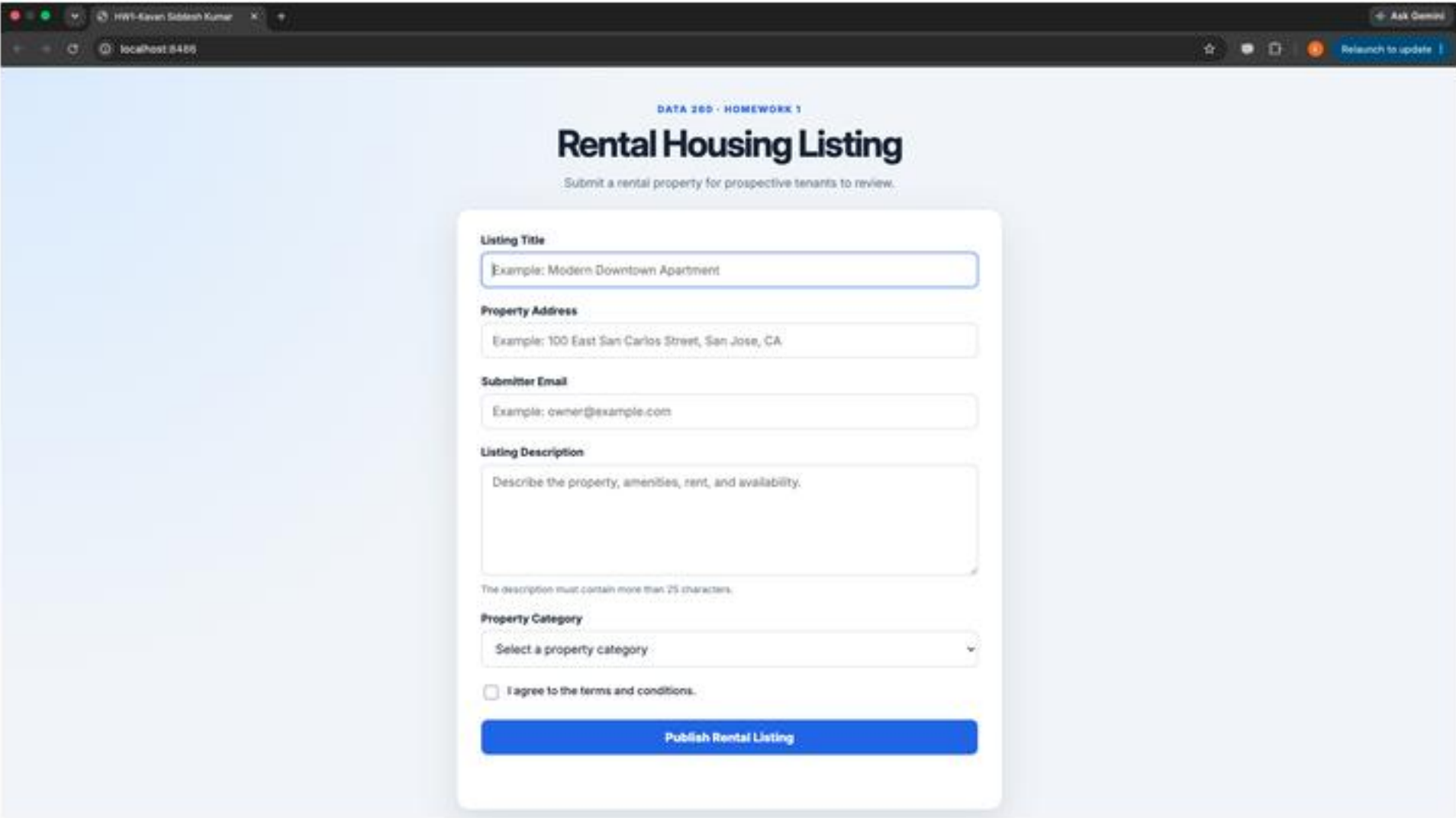
Local app in the browser



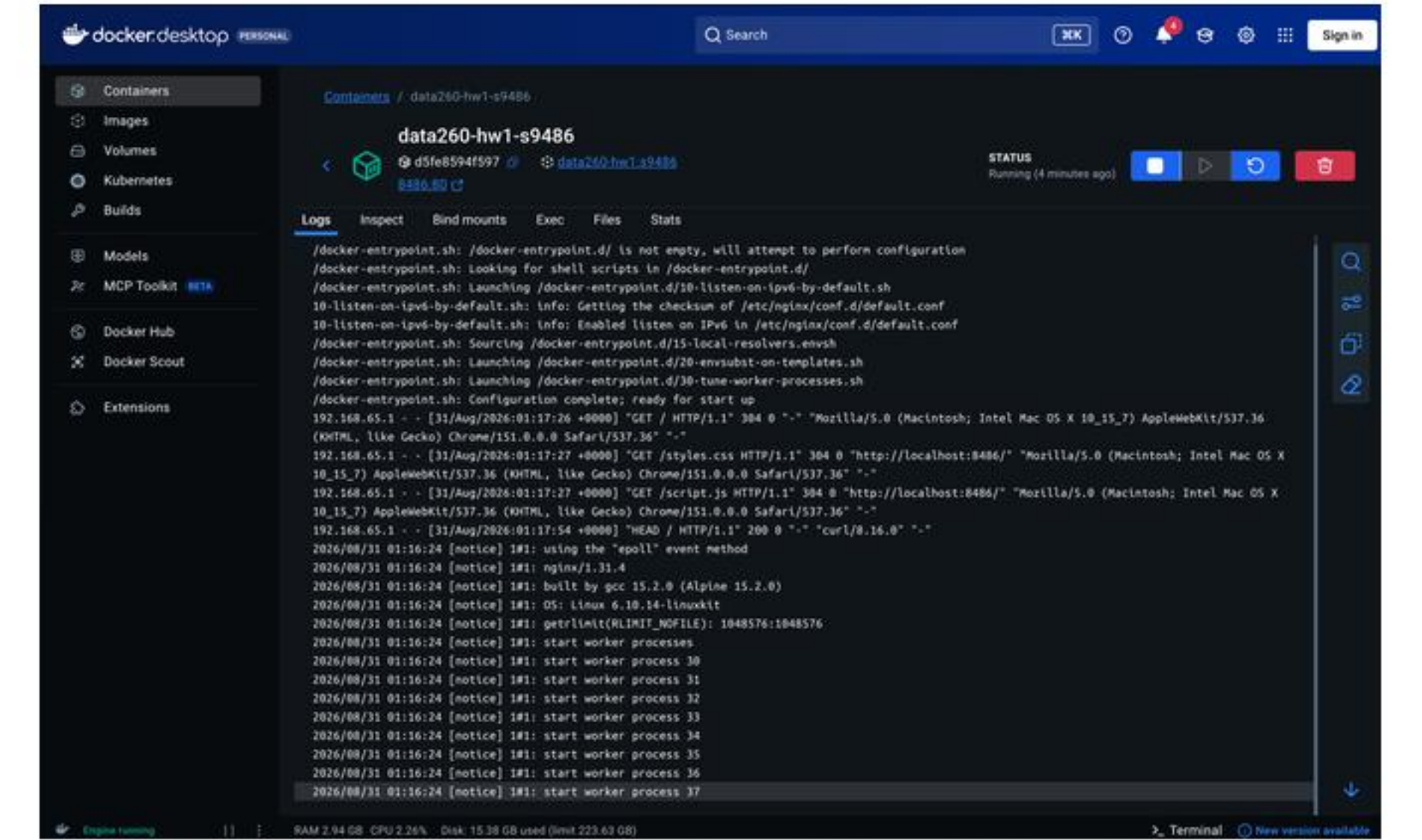
Invalid description submission



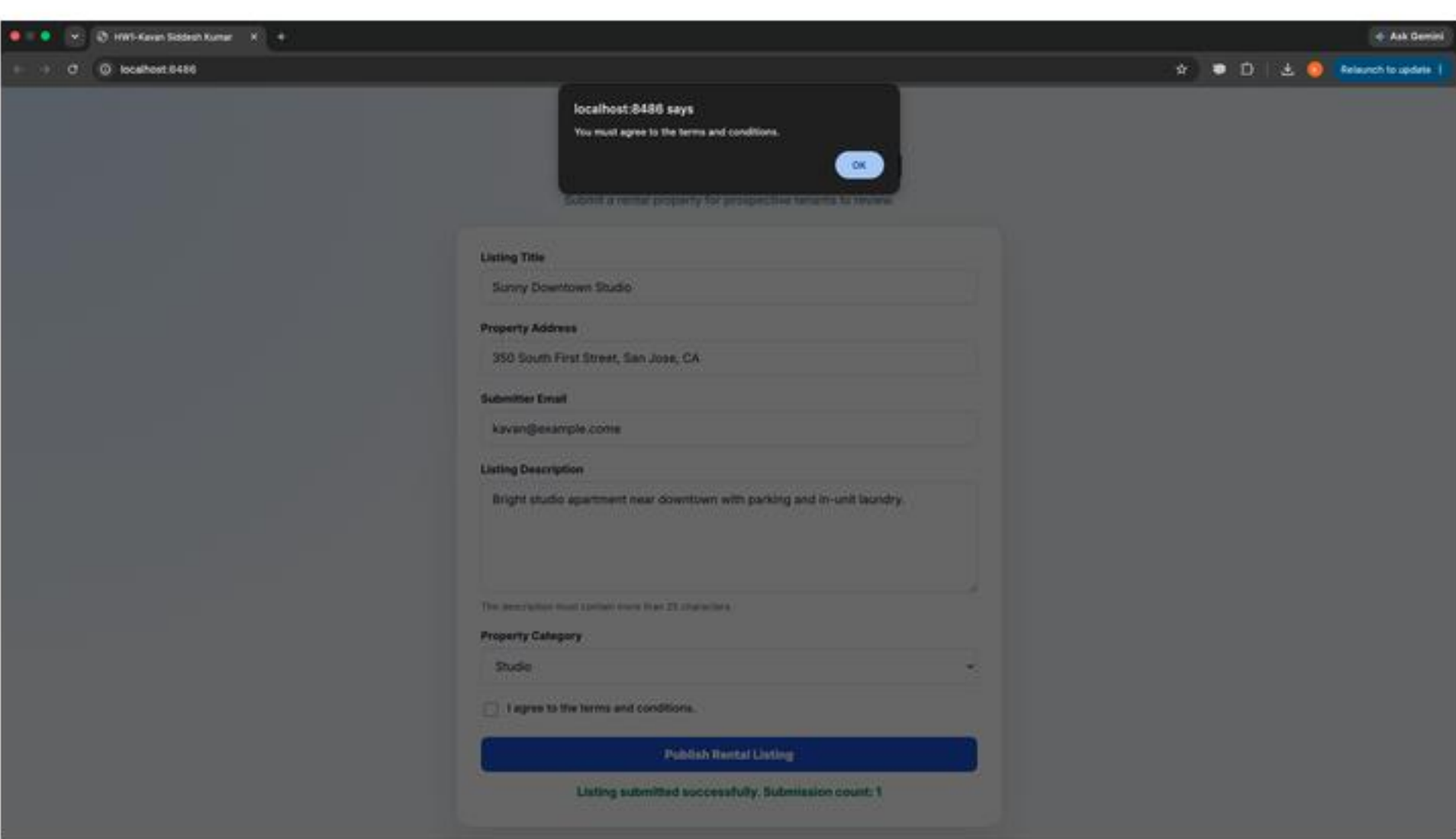
Missing terms submission



Docker-backed localhost app



Docker container evidence

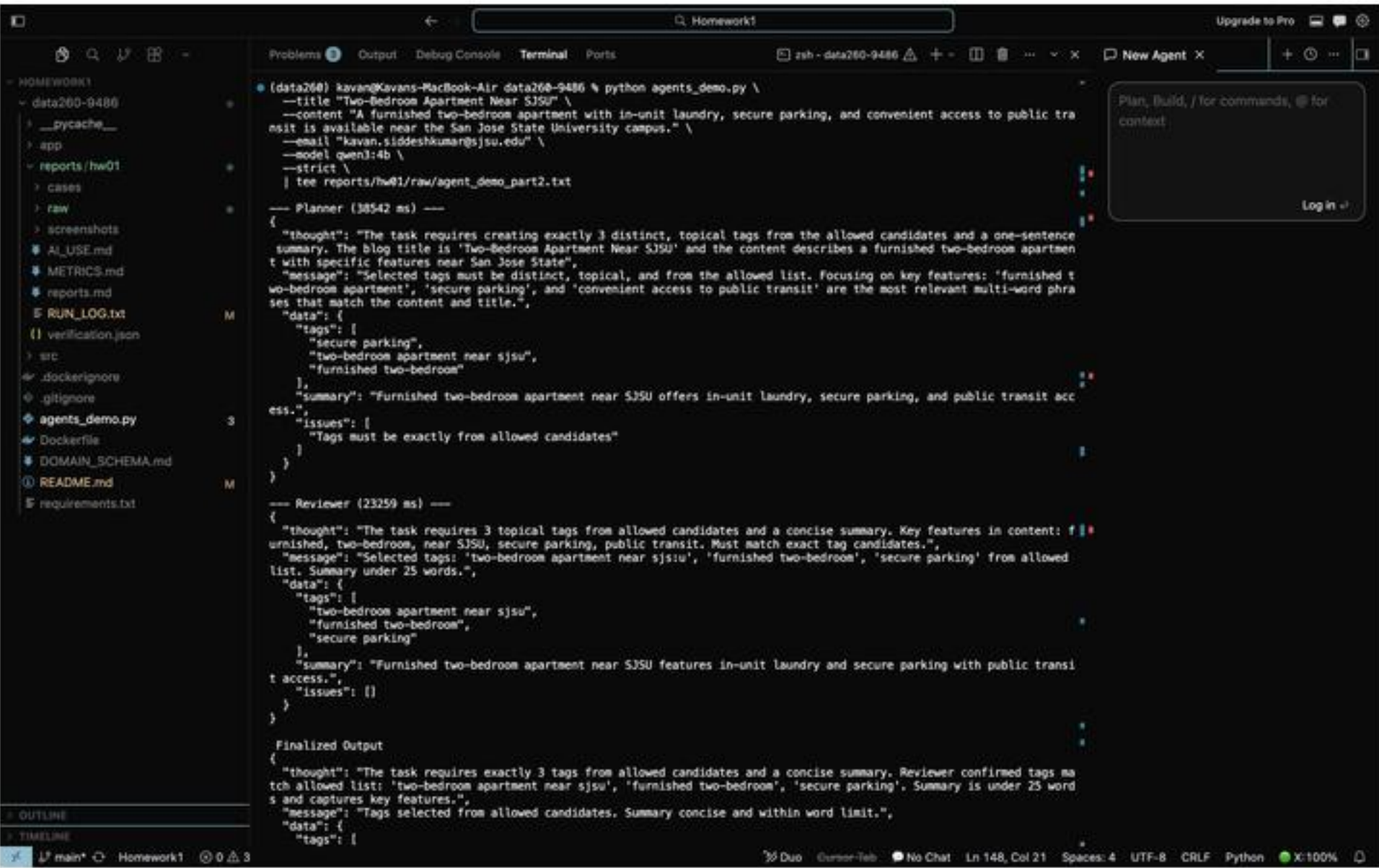


ECS public IP evidence

Section 2. Part 2: Multi-Agent Rental-Domain Demo

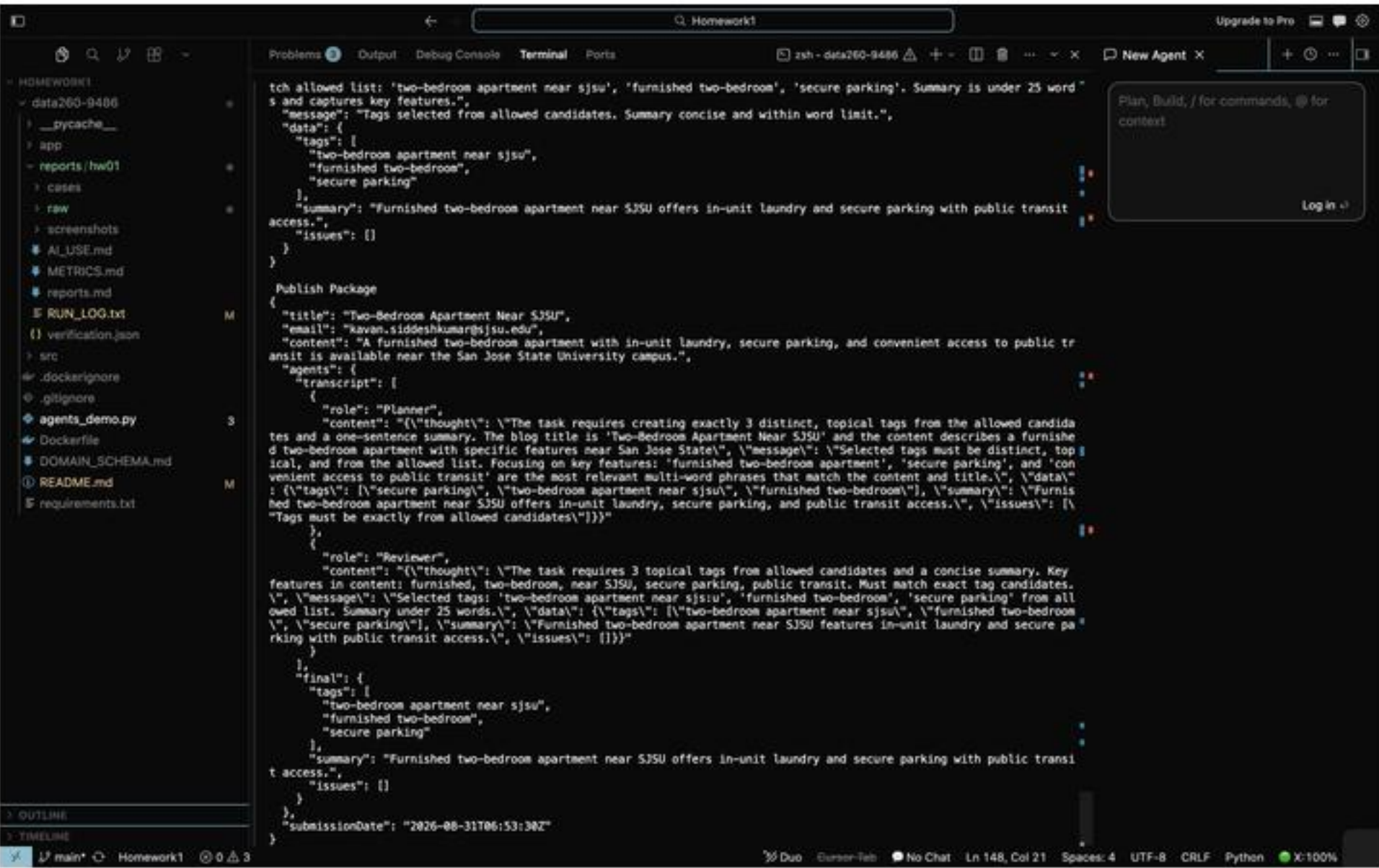
Exact command

```
python3 agents_demo.py --title "Two-Bedroom Apartment Near SJSU" --content "A furnished two-bedroom apartment with in-unit laundry, secure parking, and convenient access to public transit is available near the San Jose State University campus." --email "kavan.siddeshkumar@sjsu.edu" --model qwen3:4b --strict | tee reports/hw01/raw/agent_demo_part2.txt
```



Planner and Reviewer

- Q1: The finalized tags were two-bedroom apartment near sjsu, furnished two-bedroom, and secure parking.
- Q2: The final summary was: Furnished two-bedroom apartment near SJSU offers in-unit laundry and secure parking with public transit access.
- Q3: No issues remained in the final output; data.issues was an empty array.



Finalized Output and Publish Package

Section 3. Part 3: Non-Determinism Measurement

The fixed experiment input is preserved in reports/hw01/cases/nondeterminism_input.json and was not regenerated.

Metric	Temperature 0.0	Temperature 0.7
Runs	20	20
Distinct tag sets	1	3
Tags in all runs	furnished two-bedroom, secure parking, two-bedroom apartment near sjsu	furnished two-bedroom
Tags in exactly one run	None	convenient access
p50 latency	74523.73 ms	80115.68 ms
p95 latency	103968.01 ms	91286.90 ms
p99 latency	120482.96 ms	96006.23 ms

HOMEWORK1

data260-9486

__pycache__

app

reports/hw01

cases

raw

.gitkeep

agent_demo_final.txt

agent_demo_part2.txt

agent_demo_strict.txt

nondeterminism_full_cons...

{ } nondeterminism_metrics.j...

nondeterminism_runs.csv

{ } nondeterminism_runs.json

nondeterminism_smoke_c...

screenshots

AI_USE.md

METRICS.md

reports.md

RUN_LOG.txt

{ } verification.json

scripts

run_nondeterminism.py

src

.dockerignore

.gitignore

agents_demo.py

Dockerfile

DOMAIN_SCHEMA.md

README.md

requirements.txt

OUTLINE

TIMELINE

nism.py

METRICS.md

{ } nondeterminism_metrics.json

{ } nondeterminism_input.json

data260-9486 > reports > hw01 > raw > { } nondeterminism_metrics.json > ...

1 {

2 "0.0": {

3 "runs": 20,

4 "distinct_tag_sets": 1,

5 "tags_in_all_runs": [

6 "furnished two-bedroom",

7 "secure parking",

8 "two-bedroom apartment near sjsu"

9],

10 "tags_in_exactly_one_run": [],

11 "latency_ms": {

12 "p50": 74523.73,

13 "p95": 103968.01,

14 "p99": 120482.96

15 }

16 },

17 "0.7": {

18 "runs": 20,

19 "distinct_tag_sets": 3,

20 "tags_in_all_runs": [

21 "furnished two-bedroom"

22],

23 "tags_in_exactly_one_run": [

24 "convenient access"

25],

26 "latency_ms": {

27 "p50": 80115.68,

28 "p95": 91286.9,

29 "p99": 96006.23

30 }

31 }

32 }

Problems

Output

Debug Console

Terminal

Ports

zsh - data260-9486

Running temperature=0.7, run=12

Completed in 83634.64 ms: ['two-bedroom apartment near sjsu', 'furnished two-bedroom', 'secure parking']

Running temperature=0.7, run=13

Completed in 73240.58 ms: ['furnished two-bedroom', 'in-unit laundry', 'two-bedroom apartment near sjsu']

Running temperature=0.7, run=14

Completed in 76277.35 ms: ['two-bedroom apartment near sjsu', 'furnished two-bedroom', 'secure parking']

Running temperature=0.7, run=15

Completed in 97186.06 ms: ['two-bedroom apartment near sjsu', 'furnished two-bedroom', 'in-unit laundry']

Running temperature=0.7, run=16

Completed in 82134.68 ms: ['two-bedroom apartment near sjsu', 'furnished two-bedroom', 'secure parking']

main*

Homework1

0 3

Duo

Cursor-Tab

No Chat

Ln 1, Col 1

Spaces: 2

UTF-8

LF

{ } JSON

X:100%

Non-determinism metrics and run output

Section 4. Part 4: Strict Code Review Client

hw1_client.py loads AGENT.md as the system prompt and sends every request through src/model_client.ModelClient.complete(messages, tools=None). The adapter centralizes the Ollama call, token accounting, and optional tool support.

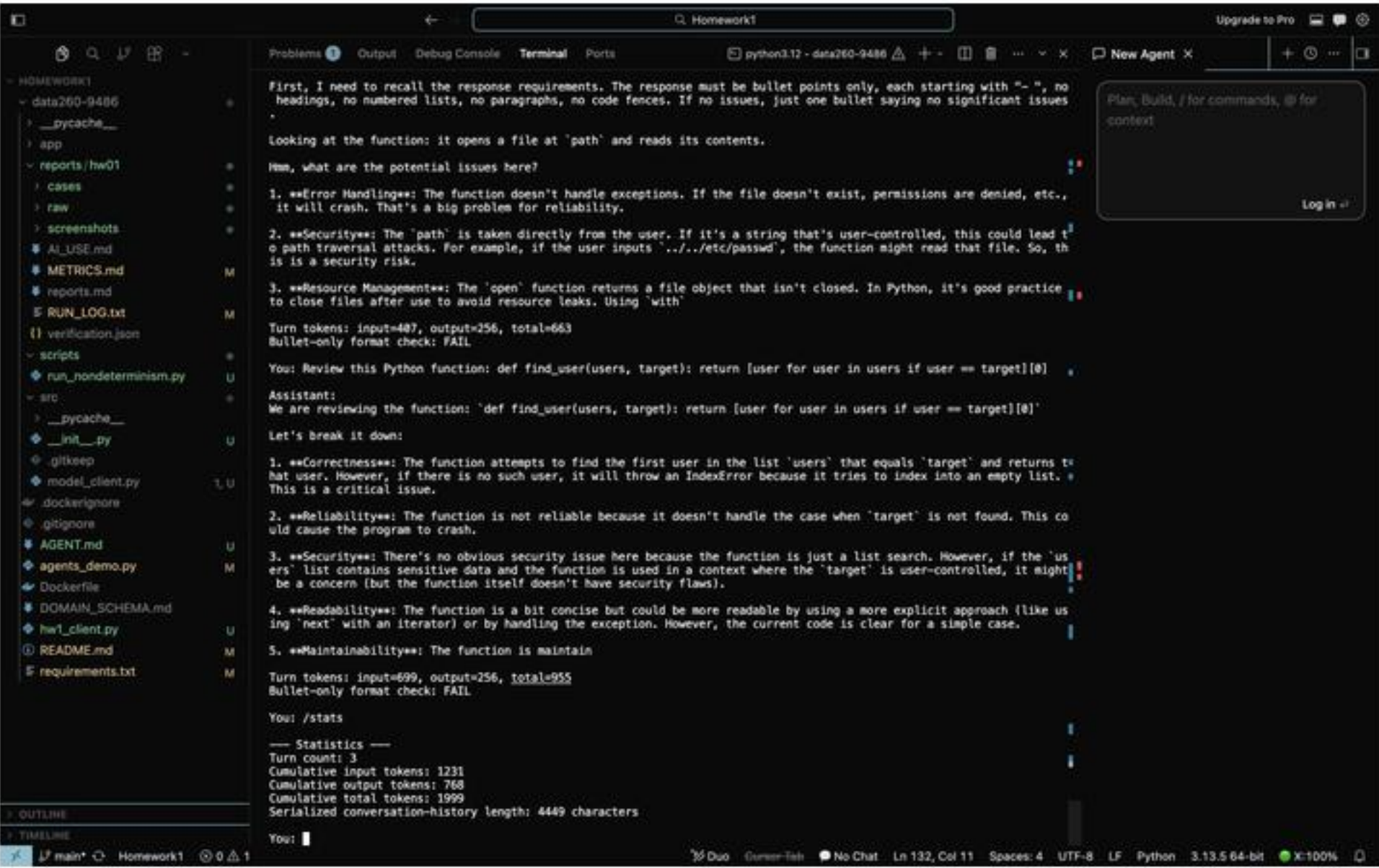
AGENT.md requires bullet-point-only code review responses.

Turn	Input tokens	Output tokens	Total tokens
1	125	256	381
2	407	256	663
3	699	256	955
4	986	256	1242
5	1272	256	1528

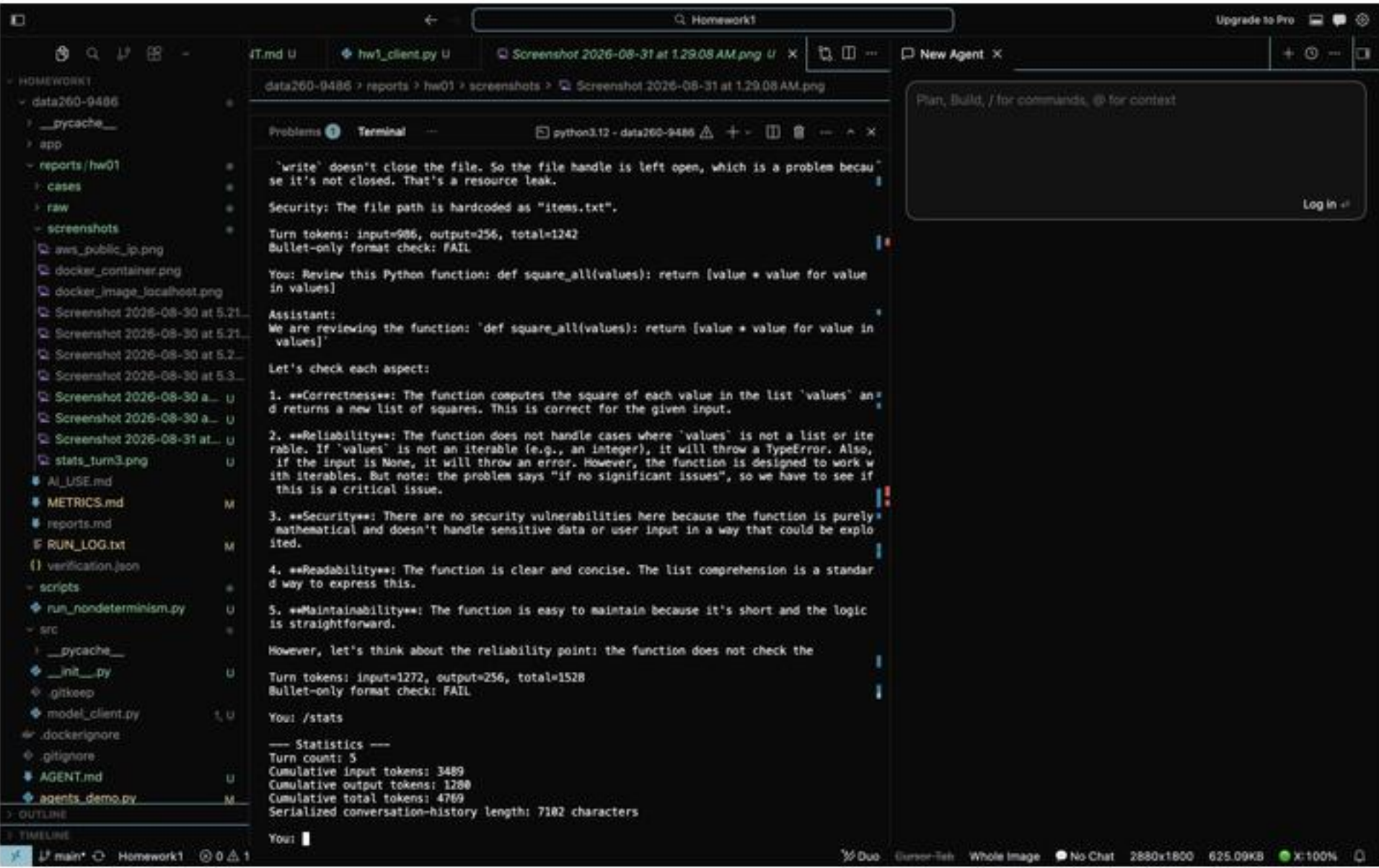
Snapshot	Turn count	Cumulative input	Cumulative output	Cumulative total	Serialized history length
After turn 3	3	1231	768	1999	4449 characters
After turn 5	5	3489	1280	4769	7102 characters

The client prints Bullet-only format check after each response. The evidence shows FAIL because the model returned numbered lists instead of pure bullet points, which confirms the checker is working and the transcript captured the non-compliant output honestly.

Section 4. Part 4 Evidence and Closing Notes



Turn 3 statistics and bullet-only check



Turn 5 statistics

Conceptual answers: prior history is resent because the chat API is stateless between requests; a system prompt sets global behavior while a user message supplies the current request; input tokens grow because earlier conversation messages are sent again on each turn; growth is eventually limited by the model context window and practical latency and memory constraints.

- AI use disclosure is recorded in reports/hw01/AI_USE.md.
- The reproducibility and verification summary lives in reports/hw01/verification.json and the self-check script scripts/verify_hw01.py.
- The final PDF for submission is reports/hw01/report.pdf.